CHAPTER QUESTIONS

SKILLS PROBLEM SOLVING

8

1 Balance the following equations:

a $Fe + HCl \rightarrow FeCl_2 + H_2$

b $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

c $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$

d $Al + Cr_2O_3 \rightarrow Al_2O_3 + Cr$

e $Fe_2O_3 + CO \rightarrow Fe + CO_2$

f $NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + CO_2 + H_2O$

g $C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$

h $Fe_3O_4 + H_2 \rightarrow Fe + H_2O$

i $Pb + AgNO_3 \rightarrow Pb(NO_3)_2 + Ag$

j $AgNO_3 + MgCl_2 \rightarrow Mg(NO_3)_2 + AgCl$

k $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

2 Calculate the relative formula masses of the following compounds:

a CO_2

b CH_3CO_2H

c Na_2SO_4

d $(NH_4)_2SO_4$

e $Na_2CO_3 \cdot 10H_2O$

f $Cr_2(SO_4)_3$

(A_r: H = 1, C = 12, N = 14, O = 16, Na = 23, S = 32, Ca = 40, Cr = 52, Fe = 56)

9

3 Calculate the percentage of nitrogen in each of the following substances (all used as nitrogen fertilisers):

a potassium nitrate, KNO_3

b ammonium nitrate, NH_4NO_3

c ammonium sulfate, $(NH_4)_2SO_4$

(A_r: H = 1, N = 14, O = 16, S = 32, K = 39)

4 Calculate the mass of the following:

a 1 mol of HCN

b 1 mol of lead(II) nitrate, $Pb(NO_3)_2$

c 4.30 mol of methane, CH_4

d 0.70 mol of Na_2O

e 0.015 mol of $NaNO_3$

f 0.24 mol of sodium carbonate crystals, $Na_2CO_3 \cdot 10H_2O$

(A_r: H = 1, C = 12, N = 14, O = 16, Na = 23, Pb = 207)

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5 Calculate the number of moles represented by each of the following:

a 20 g of magnesium oxide, MgO

b 3.20 g of iron(III) oxide, Fe_2O_3

c 2 kg of copper(II) oxide, CuO

d 50 g of copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$

e 1 tonne of iron, Fe (1 tonne is 1 000 000 g)

f 0.032 g of sulfur dioxide, SO_2

(A_r: H = 1, O = 16, S = 32, Mg = 24, Fe = 56, Cu = 63.5)

6 The following calculations use A_r: H = 1, O = 16, Na = 23, Cl = 35.5, Ca = 40, Cu = 63.5.

a Calculate the mass of 4 mol of sodium chloride, NaCl.

b Calculate how many moles is 37 g of calcium hydroxide, $Ca(OH)_2$.

c Calculate how many moles is 1 kg (1000 g) of calcium, Ca.

d Calculate the mass of 0.125 mol of copper(II) oxide, CuO.

e 0.1 mol of a substance has a mass of 4 g. Calculate the mass of 1 mole.

f 0.004 mol of a substance has a mass of 1 g. Calculate the relative formula mass of the compound.

7 Determine the empirical formulae of the compounds which contain:

a 9.39 g P, 0.61 g H

b 5.85 g K, 2.10 g N, 4.80 g O

c 3.22 g Na, 4.48 g S, 3.36 g O

d 22.0% C, 4.6% H, 73.4% Br (by mass)

(A_r: H = 1, C = 12, N = 14, O = 16, Na = 23, S = 32, K = 39, Br = 80)

8 1.24 g of phosphorus was burned completely in oxygen to give 2.84 g of phosphorus oxide. Find:

a the empirical formula of the oxide

b the molecular formula of the oxide given that 1 mole of the oxide has a mass of 284 g.

(A_r: O = 16, P = 31)

9 An organic compound contained C 66.7%, H 11.1%, O 22.2% by mass. Its relative formula mass was 72. Find:

a the empirical formula of the compound

b the molecular formula of the compound.

(A_r: H = 1, C = 12, O = 16)

10 In an experiment to find the number of molecules of water of crystallisation in sodium sulfate crystals, $Na_2SO_4 \cdot nH_2O$, 3.22 g of sodium sulfate crystals were heated gently. When all the water of crystallisation had been driven off, 1.42 g of anhydrous sodium sulfate was left. Find the value of n in the formula. (A_r: H = 1, O = 16, Na = 23, S = 32)

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11 Gypsum is hydrated calcium sulfate, $CaSO_4 \cdot nH_2O$. A sample of gypsum was heated in a crucible until all the water of crystallisation had been driven off. The following results were obtained:

Mass of crucible = 37.34 g

Mass of crucible + gypsum, $CaSO_4 \cdot nH_2O$ = 45.94 g

Mass of crucible + anhydrous calcium sulfate, $CaSO_4$ = 44.14 g

Calculate the value of n in the formula $CaSO_4 \cdot nH_2O$.

(A_r: H = 1, O = 16, S = 32, Ca = 40)

12 **a** Calculate the amount in moles of SO_3 formed when 0.36 mol SO_2 reacts with excess O_2.

$2SO_2 + O_2 \rightarrow 2SO_3$

b Calculate the amount in moles of HCl that reacts with 0.4 mol $CaCO_3$.

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

c Calculate the amount in moles of H_2S formed when 0.4 mol of HCl reacts with excess Sb_2S_3.

$Sb_2S_3 + 6HCl \rightarrow 2SbCl_3 + 3H_2S$

d Calculate the amount in moles of iron formed when 0.9 mol carbon monoxide reacts with excess iron(III) oxide.

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

e Calculate the amount in moles of hydrogen that would be required to make 0.8 mol NH_3.

$N_2 + 3H_2 \rightarrow 2NH_3$

13 The reaction between iron and bromine is

$2Fe + 3Br_2 \rightarrow 2FeBr_3$

10 g of iron is reacted with bromine.

a Calculate the amount in moles of iron that reacted.

b Calculate the amount in moles of bromine that the iron reacted with.

c Calculate the amount in moles of $FeBr_3$ that will be formed.

d Calculate the mass of $FeBr_3$ formed.

(A_r: Fe = 56, Br = 80)

14 Titanium is manufactured by heating titanium(IV) chloride with sodium.

$TiCl_4(g) + 4Na(l) \rightarrow Ti(s) + 4NaCl(s)$

1.0 g of $TiCl_4$ is reacted with excess sodium

(A_r: Na = 23, Ti = 48, Cl = 35.5)

a Calculate the amount in moles of $TiCl_4$ reacted.

b Calculate the amount in moles of Ti formed.

c Calculate the mass of Ti formed.

d Calculate the mass of NaCl formed.

e Calculate the mass of Ti formed when 1 tonne of $TiCl_4$ reacts with excess sodium (1 tonne is 1 000 000 g).

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15 2.67 g of aluminium chloride was dissolved in water and an excess of silver nitrate solution was added to give a precipitate of silver chloride (AgCl).

$AlCl_3(aq) + 3AgNO_3(aq) \rightarrow Al(NO_3)_3(aq) + 3AgCl(s)$

Calculate the mass of silver chloride that would be formed.

(A_r: Al = 27, Cl = 35.5, Ag = 108)

16 Chromium is manufactured by heating a mixture of chromium(III) oxide with aluminium powder.

$Cr_2O_3(s) + 2Al(s) \rightarrow 2Cr(s) + Al_2O_3(s)$

a Calculate the mass of aluminium needed to react with 50 g of Cr_2O_3.

b Calculate the mass of chromium produced from 50 g of Cr_2O_3.

c Calculate the mass of chromium produced from 5 kg of Cr_2O_3.

d Calculate the mass of chromium produced from 5 tonnes of Cr_2O_3.

(1 tonne is 1 000 000 g)

(A_r: Cr = 52, O = 16, Al = 27)

17 Copper(II) sulfate crystals, $CuSO_4 \cdot 5H_2O$, can be made by heating copper(II) oxide with dilute sulfuric acid and then crystallising the solution formed.

a Calculate the maximum mass of crystals that could be made from 4.00 g of copper(II) oxide using an excess of sulfuric acid.

$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$

$CuSO_4(aq) + 5H_2O(l) \rightarrow CuSO_4 \cdot 5H_2O(s)$

b If the actual mass of copper(II) sulfate collected at the end of the experiment was 11.25 g, calculate the percentage yield.

(A_r: H = 1, O = 16, S = 32, Cu = 63.5)

18 Ethanol reacts with ethanoic acid to form ethyl ethanoate and water:

$C_2H_5OH(l) + CH_3COOH(l) \rightarrow CH_3COOC_2H_5(l) + H_2O(l)$

ethanol ethanoic acid ethyl ethanoate water

a Calculate the mass of ethyl ethanoate produced when 20.0 g of ethanol reacts with excess ethanoic acid.

b The actual yield in this experiment is 30.0 g of ethyl ethanoate. Calculate the percentage yield.

(A_r: C = 12, H = 1, O = 16)

SKILLS REASONING

19 Determine which reactant is in excess in each case.

a 0.5 mol of Na_2CO_3 is reacted with 0.5 mol hydrochloric acid:

$Na_2CO_3(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

b 0.01 mol C_3H_8 is reacted with 0.02 mol O_2:

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

c 28 g CO is reacted with 0.6 mol Fe_2O_3:

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

d 16 g O_2 is reacted with 16 g SO_2:

$2SO_2 + O_2 \rightarrow 2SO_3$

(A_r : C = 12, O = 16, S = 32)

20 1.0 g of $CaCO_3$ is reacted with 0.015 mol of HCl:

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

a Determine which reagent is present in excess.

b Calculate the mass of CO_2 produced.

(A_r: Ca = 40, C = 12, O = 16)

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